$$E_1 = \frac{7}{4}\frac{ml^2}{T^2} \qquad E_2 = \frac{1}{4}\frac{ml^2}{T^2} \qquad E_3 = \frac{17}{36}\frac{ml^2}{T^2} \tag{9}$$

$t = T/2$ C_1 $u = 0$

C_1 v_n'

$$\frac{l}{v_n'} = \frac{n}{2}T \qquad n = 1,2,\cdots \tag{10}$$

C_2

$$v_n' = \frac{2l}{nT} \qquad n = 1,2,\cdots \tag{11}$$

C_1 C_2 C_2 G

$$\Delta t_{n1}' = \frac{n}{2}T \qquad \Delta t_{n2}' = \frac{n}{4}T \tag{12}$$

G

$$t_n' = \frac{T}{2} + \Delta t_{n1}' + \Delta t_{n2}' = \frac{3n+2}{4}T \tag{13}$$

D_2 $t = 10T/4$

$$t_n' \leqslant \frac{10}{4}T \tag{14}$$

(13) (14) n

$$n_{\max} = 2 \tag{15}$$

$$E_n' = \frac{1}{2}mv_2'^2 + E_0\sin\left[\left(\frac{2\pi}{T}\frac{3n+2}{4}\right)T + \frac{\pi}{4}\right]$$
$$n = 1,2 \tag{16}$$

$$E_1' = \frac{9}{4}\frac{ml^2}{T^2} \qquad E_2' = \frac{3}{4}\frac{ml^2}{T^2} \tag{17}$$

D_1 $t = 3T/4$ C_1 (9)、(17) E_1'

V_1'

$$E_1' = \frac{1}{2}MV_1'^2$$

$$V_1' = \sqrt{\frac{9}{2}}\frac{l}{T} \tag{18}$$

E_2 V_2

$$V_2 = \sqrt{\frac{1}{2}}\frac{l}{T} \tag{19}$$

2.

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